

Other than review your past homework and pop-quiz, here are some questions that will also show up in mid-term:

- 1) Taylor series! How Euler's formula is derived via Taylor series expansion.
- 2) Solve unforced ODE (like a harmonic oscillator) with good guessing ($e^{\lambda t}$ or $\cos(\omega t + \varphi)$), or maybe prove that they can yield the same solution ...
- 3) Solve 2nd order ODE with eigenvalue and eigenvector, given initial conditions of course...
- 4) Potential energy and conservative force! You could find conservative force from any potential field with this simple operator...
- 5) Knows how to derive phase portrait of a bead sliding on a weird potential field by:
 - Guessing the fixed point;
 - Get the eigenvalues of the linearization around each point;
 - Consider both no damping (no friction) and damping (with friction) conditions.
- 6) Derive equation of motion with coupled oscillators, find it's mode and frequency by the eigenstuff, and I can tell you that you should review how you "det" a (3x3) matrix.
- 7) Use Lagrange mech to derive equation of motion like example 7.3 (Atwood machine) and 7.5 (Block sliding on a wedge) from Taylor's textbook.